Project Planning Phase

Date	05 AUG 2026
Student Name	P. Srujan Sinha
Project Name	Intergalactic Space Travel Request (ISTR) System
Maximum Marks	4 Marks

1. Agile Planning Concepts

The ISTR project was planned using standard Agile/Scrum concepts, applied as follows:

Sprint

A fixed period of time (here, 6 days) in which the team commits to completing a defined set of tasks.

Epic

A large body of work — such as “Approval Workflow” or “Security & Access Control” — too big to complete in a single sprint, broken down into smaller stories.

Story

A small, deliverable unit of work belonging to an Epic, written from the user's perspective (e.g. “As a Manager, I can approve a mission request”).

Story Point

A relative measure of the effort required to complete a story, estimated using the Fibonacci-like scale below:

Story Points	Effort Level
1	Very Easy task
2	Normal task
3	Moderate task
5	Difficult task

2. Product Backlog, Sprint Schedule & Estimation

The ISTR backlog is organized into 5 Epics across 3 sprints, covering all 10 functional modules of the application.

Sprint	Epic (Functional Req.)	US No.	User Story / Task	Story Points	Priority	Team
Sprint-1	Mission Request Management	USN-1	As a Mission Requester, I can submit a mission request with destination, travel dates, crew details, and budget.	3	High	Srujan
Sprint-1		USN-2	As a Mission Requester, I can save a request as Draft before final submission.	2	Medium	Srujan

Sprint	Epic (Functional Req.)	US No.	User Story / Task	Story Points	Priority	Team
Sprint-1		USN-3	As a Mission Requester, I can attach supporting mission documents to my request.	2	Medium	Srujan
Sprint-1	Approval Workflow	USN-4	As a Manager, I can review and approve or reject a submitted mission request.	3	High	Srujan
Sprint-1		USN-5	As a Finance Officer, I can approve the budget for missions flagged for financial review.	3	High	Srujan
Sprint-2	Approval Workflow	USN-6	As a Mission Control Officer, I can give final approval and schedule the mission.	3	High	Srujan
Sprint-2	Security & Access Control	USN-7	As a System Administrator, I can configure roles and groups for each department.	2	Medium	Srujan
Sprint-2		USN-8	As a System Administrator, I can create ACLs to secure record access by role.	3	High	Srujan
Sprint-2	Dynamic Forms	USN-9	As a Mission Requester, I see the Diplomatic Officer field only when Mission Type is Alien Negotiation.	2	Medium	Srujan
Sprint-3	Notifications	USN-10	As a Mission Requester, I receive automated emails when my request is submitted, approved, rejected, or completed.	3	High	Srujan
Sprint-3	Dashboard	USN-11	As a Manager, I can view a Mission Analytics dashboard showing pending, approved, and rejected requests.	5	High	Srujan
Sprint-3	Reports	USN-12	As a System Administrator, I can generate Mission Status, Approval Summary, and Risk Analysis reports.	3	Medium	Srujan

3. Sprint-wise Story Point Calculation

Sprint 1

- Mission Request Management: USN-1 (3) + USN-2 (2) + USN-3 (2) = 7
- Approval Workflow: USN-4 (3) + USN-5 (3) = 6

Total Story Points in Sprint 1 = 7 + 6 = 13

Sprint 2

- Approval Workflow: USN-6 (3) = 3
- Security & Access Control: USN-7 (2) + USN-8 (3) = 5
- Dynamic Forms: USN-9 (2) = 2

Total Story Points in Sprint 2 = 3 + 5 + 2 = 10

Sprint 3

- Notifications: USN-10 (3) = 3
- Dashboard: USN-11 (5) = 5
- Reports: USN-12 (3) = 3

Total Story Points in Sprint 3 = 3 + 5 + 3 = 11

Overall Velocity

Total Story Points = 13 + 10 + 11 = 34

Number of Sprints = 3

Velocity = Total Story Points ÷ Number of Sprints = 34 ÷ 3 ≈ 11.3

The ISTR team's velocity is approximately 11.3 Story Points per Sprint.

4. Project Tracker, Velocity & Burndown Chart

Sprint	Total SP	Duration	Start Date	End Date (Planned)	SP Completed	Release Date (Actual)
Sprint-1	13	6 Days	01 Jun 2026	06 Jun 2026	13	06 Jun 2026
Sprint-2	10	6 Days	08 Jun 2026	13 Jun 2026	10	13 Jun 2026
Sprint-3	11	6 Days	15 Jun 2026	20 Jun 2026	11	20 Jun 2026

Velocity

With a 6-day sprint duration and a team velocity of 11.3 story points per sprint, the average velocity per day works out to approximately 11.3 ÷ 6 ≈ 1.9 story points per day.

Burndown Chart

A burndown chart tracks remaining work (story points) against time, comparing the ideal completion rate to the team's actual progress across all three sprints of the ISTR project.

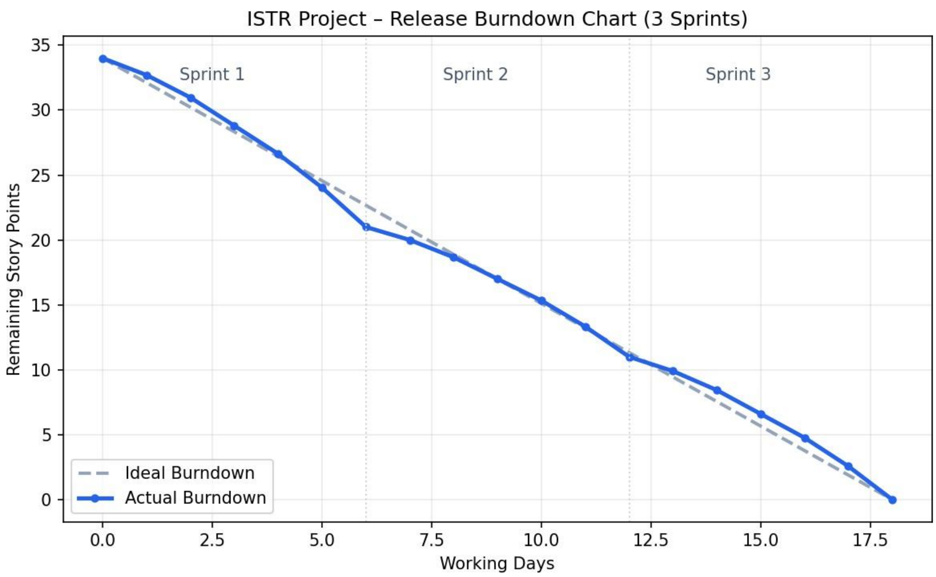

Figure 1: ISTR Release Burndown Chart (Sprint 1–3)

References

1. Atlassian – Agile Project Management: atlassian.com/agile/project-management

2. Atlassian – Epics: atlassian.com/agile/tutorials/epics

3. Atlassian – Sprints: atlassian.com/agile/tutorials/sprints

4. Atlassian – Estimation: atlassian.com/agile/project-management/estimation

5. Atlassian – Burndown Charts: atlassian.com/agile/tutorials/burndown-charts